A Framework for Forecasting Professional Tennis Matches Using ATP Statistics and Betting Odds

Lorenzo Trucco[a], Domagoj Tolić[b]
[a]*RIT Croatia, Ul. Damira Tomljanovića 15, 10000 Zagreb, Croatia*
[b]*RIT Croatia, Don Frana Bulića 6, 20000 Dubrovnik, Croatia*
let2966@rit.edu

***Abstract*—We propose a machine learning framework to forecast professional tennis outcomes using a 57,000-match dataset (2003–2024). By combining engineered ATP statistics with pre-match betting odds, the model captures both player dynamics and market expectations. Comparative results show that ATP-based features alone achieve performance within 1-2% of odds-only models, confirming the power of domain-driven engineering. Our final XGBoost model reaches nearly 70% accuracy on unseen matches using only pre-match data.**

***Index Terms*—Machine learning, Tennis prediction, Sports analytics, XGBoost, Feature engineering, Betting odds**

I. Introduction

The main intention of this work is to determine whether it is possible for Machine Learning (ML) to provide a base for finding a statistical edge in sports betting, and more so in a real-world setting [1], [2]. This study is purely statistical and detached from gambling; in fact, its purpose is to examine the effectiveness of prediction models in a controlled context and not to define a betting strategy. One of the most important aspects of this research is the strict methodology followed in the data collection processes, feature engineering, and model construction to render the framework more realistic.

The present study integrates market data in the form of betting odds, thereby capturing instantaneous probabilistic expectations. Since contemporary betting companies are multinational corporations that draw on the wealth of experience of sophisticated data scientists and analysts, the odds published by these companies represent carefully streamlined probability estimates. Our original model was constructed solely on the basis of performance data and tournament results, but the subsequent inclusion of betting odds helped the model to become even better. Tennis is selected because it is more controlled and predictable than team sports (i.e., one-to-one competitions with fewer variables that determine the winner).

While the integration of bookmaker odds does improve predictive accuracy, we find that engineered Association of Tennis Professionals (ATP) features alone offer highly competitive performance – within 1-2 percentage points of odds-only models. This supports the idea that careful feature construction, informed by domain knowledge, can nearly match the predictive power of market expectations derived from experts. Our final model, which combines both data sources, reaches nearly 70% accuracy and approaches the commonly reported 72% benchmark [3], [4], despite relying on a relatively simple XGBoost algorithm. These results underscore the value of intelligent data engineering in producing robust and interpretable sports forecasting models.

The contributions of this work are:

- **Machine Learning Model Development:** An XGBoost-based predictive framework that is highly effective and efficient in structured data problems.
- **Integration of Betting Market Data:** Unlike conventional methods that purely utilize historical match data, this research integrates *pre-match* odds, thus encapsulating the probabilistic expectations of bookmakers in real time.
- **Feature Engineering and Realism:** The model ensures that it only uses *pre-match* features, preventing data leakage and aligning the study with real-world betting constraints.
- **Results and Benchmarking:** Our predictive model achieves an overall prediction accuracy of approximately 70%, highly comparable to other more advanced ML techniques such as LightGBM and Support Vector Machines (SVM) [5].

The structure of the paper is as follows. Section 2 reviews existing literature on ML applications in sports predictions whilst Section 3 outlines the datasets used in our study (i.e., historical results and bookmaker odds) along with a description of the preprocessing techniques used to maintain data integrity. Section 4 describes our modeling approach including XGBoost, the development of relevant features, hyperparameter tuning using a grid search, and the validation system for our prediction model. Section 5 brings numerical results and comparisons. Section 6 presents concluding remarks.

II. Literature Review

The implementation of ML algorithms, particularly Extreme Gradient Boosting (XGBoost), has gained prominence in sports analytics due to their predictive power and efficiency. In association football, [6] develops an improved expected goals (xG) model by incorporating event sequences leading up to a shot. Similarly, in the context of the National Basketball Association (NBA), [7] integrates XGBoost with SHapley Additive exPlanations (SHAP) to predict game outcomes. This approach not only provides accurate predictions but also offers interpretability by identifying key performance indicators such as field goal percentage, defensive rebounds, and turnovers.

Tennis constitutes a solid case for the use of ML technologies due to its organized structure and the relatively small number of external variables influencing results, unlike team sports. In the contemporary context, [8] investigates the possibility of applying supervised learning techniques to predict the outcomes of professional tennis matches using player statistics and match conditions as underlying factors. That research applies main performance indicators such as serve points won, return points won, and break points saved in developing a logistic regression model. The results indicate that statistical metrics such as break-point conversion ratio and effectiveness are great predictors of match outcomes. In a similar way, [4] investigates ML techniques to professional tennis match prediction, contrasting several models to compare their predictive capacity. That research suggests the possibility of more advanced feature engineering techniques and non-traditional model architectures outside of the standard regression-based models. The authors of [3] investigate the use of the Glicko rating system – a Bayesian approach originally designed for chess – to predict the results of Grand Slam tennis matches. The findings indicate that professional tennis rankings are the most significant predictor of match outcomes, and the Glicko-based model achieved slightly higher accuracy compared to traditional methods.

Several studies employ traditional ML algorithms to predict match outcomes with notable success. For example, Zhang (2024) uses a Random Forest model to predict players' scores in tennis matches, achieving an accuracy of approximately 83%. That study also identified serve strength as one of the most important predictive features [9]. Other research efforts have applied more conventional techniques such as LightGBM, Support Vector Machines (SVM), and Linear Regression, reaching prediction accuracies around 70%. The study [5] conducts a comparative study focusing on the prediction of tennis players' performance using time series models.

It is important to note that the above models often rely on extensive datasets, including live or in-match statistics, detailed statistics or enriched categorical variables (and hence the success rate above 80%). In contrast, our approach intentionally limits itself to a more streamlined and interpretable set of characteristics, excluding live match data and focusing exclusively on pre-match information. Essentially, our findings confirm the viability of classical models in tennis forecasting and provide a useful benchmark for comparing the performance of more recent approaches.

III. Dataset

We use the following two datasets to predict tennis match outcomes:

1. ATP Match Results (1991-2024)

The first data set comes from [10] covering **ATP matches from 1991 to 2024**. It contains detailed match information, including tournament details, player statistics, and performance metrics. This data set originally contained **105,162 matches** over the 33-year period.

2. Tennis Betting Odds (2003-2024)

The second data set originates from [11], covering **2003 to 2024**. This data set contains not only the results of the matches, but also the betting odds **before the match** from major sports books. This dataset contains **57,923 matches**, covering **fewer tournaments per year** compared to the ATP results data set.

Data Merging Process

To integrate the two datasets, we performed an **inner join**, keeping only matches present in both sources [12]. Due to naming inconsistencies, we applied **fuzzy matching** based on **Levenshtein distance** [13], with **manual corrections** for edge cases (e.g., `"R. Nadal"` vs `"Rafael Nadal"`). The final merged dataset includes **57,765 matches**, successfully linking most betting records to their corresponding ATP entries.

The Importance of Odds Data

The inclusion of betting odds serves a crucial purpose. Although traditional match data provide **historical player performance insights**, odds reflect **real-time market expectations**, integrating external factors such as injuries, momentum and public perception. This approach allows us to explore how market trends align (or deviate) from actual match outcomes, providing insights into optimization inefficiencies that could be used for prediction.

IV. Methodology

A. Data Architecture Overview

Determining the most significant variables is a crucial task in the construction of a predictive model. A starting analysis suggests that player rank is a significant predictor of match outcome (as in [3]), but several other variables significantly influence performance and play style. These include:

- **Surface Type**: Different surfaces (e.g., hard, clay, grass) impact movement, shot selection, and endurance;
- **Tournament Importance**: Prestige and point allocation affect player motivation and consistency;
- **Player Momentum**: Recent form (win/loss streaks) indicates confidence and fitness;
- **Betting Odds**: Reflect market expectations and embed predictive signals;
- **Match Context**: Finals or rivalries may influence behavior;
- **Playing Style**: Players tend to be defensive or aggressive.

To refine feature selection, a correlation analysis was conducted to detect relationships and reduce multicollinearity.

B. Target Definition and Dataset Balancing

The binary target variable indicates whether `Player 1` won (`target` = 1) or lost (`target` = 0). Since the original dataset included separate columns for winners and losers, we applied a randomization strategy: in 50% of the cases, the roles of winner and loser were swapped. This ensured an even distribution of outcomes and eliminated structural bias.

By doing so, the model learns patterns from both sides and no additional balancing techniques, such as SMOTE [14] or class weighting, were necessary. Robustness in evaluation is ensured through metrics like F1-score and AUC-ROC [15].

C. Feature Engineering

Feature engineering is key to building predictive models. It involves transforming raw match and player statistics into meaningful variables that capture trends in performance, form, and context. Tennis literature and domain knowledge guide this process.

1) Core Predictors: Ranking, Odds and Surface-Specific Performance: Ranking and bookmaker odds are among the strongest predictors in tennis forecasting, as they reflect reputation and market expectations. However, raw ranking differences tend to dominate due to their scale and skewness. To address this, a **logarithmic transformation** was applied, improving distributional symmetry (Figure 1). Additional features include `rank_pts_diff` and a composite `dominance_index` based on both rankings and odds.

Following [4], surface-specific metrics were introduced to reflect contextual effectiveness. These include each player's win rate on the current surface (`P1_Surface_Win_Rate`, `P2_Surface_Win_Rate`), their difference (`win_rate_su_diff`), and a weighted version prioritizing deeper tournament rounds.

Regarding betting odds, the most informative representation was the **implied probability**, computed as the inverse of the odds. Features like `implied_prob_p1`, `implied_prob_p2`, and their difference (`odds_diff_prob`) were found to be especially predictive (see Table IV).

2) Tournament Context: Inspired by [16], we integrate tournament-level context to better capture performance dynamics. This includes `match_spw` (total serve points won by both players) and contextual features such as:

- **tourney_importance**: prestige of the tournament (e.g., Grand Slam = 5, Challenger = 1),
- **round_weight**: custom round weights (e.g., final = 8),
- **match_weight**: interaction of tournament importance and round weight.

Additionally, we introduce the **Tournament Momentum Index (TMI)**, a dynamic feature updated match-by-match based on game margin, opponent odds, and round importance. This captures intra-tournament momentum consistently; TMI is computed separately for each player.

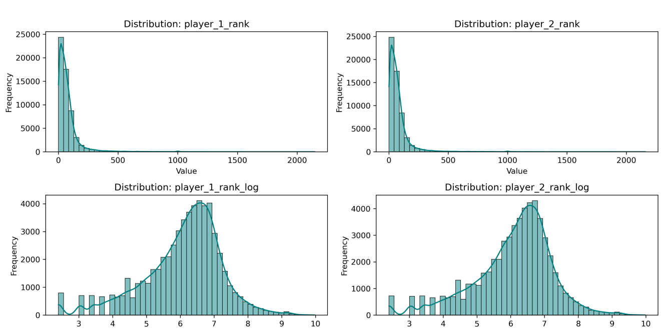

Fig. 1. Distributions of raw and log-transformed ranking variables. Log transformation improves symmetry and reduces skewness.

3) Momentum and Recent Form: To capture recent trends in player performance, we compute momentum as the difference between short-term (last 5 matches) and long-term (last 50 matches) rolling averages.

Aggressiveness Momentum and **Defense Momentum** are based on serve-related statistics such as ace rate, double fault rate, and win percentages on first and second serve. We also include a **Set Finalization Momentum**, reflecting the average margin of games per set, and an **Odds Momentum**, defined as the difference between a 15-match rolling average of betting odds and the current odds. These indicators help the model track a player's recent form and shifts in perceived strength over time.

4) Modeling Player-Specific Characteristics: Beyond standard metrics like rankings or surface preferences, several features were engineered to capture individual player traits: aggressiveness (ace-to-double fault ratio), defensive solidity (service resilience stats), mental toughness (performance in tie-breaks and deciders), set dominance (average margin per set), stylistic matchups (aggressiveness vs opponent defense), and upset potential from wildcard low-ranked entries.

D. Feature Selection Strategy

An iterative feature selection process is adopted next. After the initial engineering of over 100 variables, exploratory data analysis (EDA) and domain-driven reasoning are used to prune irrelevant or redundant features. Multicollinearity is tested using correlation matrices, and variables with high redundancy are excluded.

E. Variable Analysis

A preliminary analysis confirms what one might theoretically expect: bookmaker odds are among the most predictive features for determining match outcomes. As they represent a synthesis of expert expectations, recent player form, contextual conditions, and public sentiment, these market-based variables are expected to exhibit strong correlations with the target.

After conducting several tests that excluded betting odds, it became evident that other feature groups also contribute meaningfully to the model's predictive power. In particular, **ranking-related variables**, emerged as the most influential among the ATP-based features. Additionally, **momentum-based indicators**—especially those capturing recent player performance fluctuations—and **surface-adjusted win rates** demonstrated notable impact. These findings suggest that, even in the absence of market-based signals, domain-specific features reflecting form, style, and surface adaptability can capture important dimensions of match dynamics.

F. Gradient Boosting

Gradient Boosting is an ML algorithm that iteratively creates a robust predictive model from an ensemble of weak learners, typically decision trees [6]. The procedure minimizes an arbitrary differentiable loss function using gradient descent. XGBoost (**eXtreme Gradient Boosting**) improves traditional Gradient Boosting by incorporating several key enhancements [17]:

- **L1/L2 Regularization:** Controls model complexity and reduces overfitting;
- **Missing Values:** Learns optimal imputation paths during training;
- **Built-in Cross-Validation:** Automatically selects the number of boosting rounds;
- **Scalability:** Parallel computation and GPU support for large datasets; and
- **Efficient Pruning:** Removes non-beneficial splits via maximum depth pruning.

Due to these optimizations, XGBoost is widely used in ML competitions and real-world applications requiring high predictive performance.

G. Grid Search Tuning

To optimize performance, we use a grid search approach [18] with a 3-fold stratified cross-validation, preserving the class distribution. The search spans key hyperparameters for the Gradient Boosting classifier, including number of estimators, learning rate, max depth, and min samples per split. The best configuration is selected based on average F1-score, ensuring robustness in presence of class imbalance and optimizing the trade-off between precision and recall. This approach enhances both model accuracy and generalizability to unseen data.

H. Evaluation Strategy

To evaluate the model's predictive performance, a chronological split of the dataset is performed, separating training, validation, and testing sets in a way that respects the temporal order of matches. This strategy mirrors real-world applications, where predictions must be made for future events based on past data.

Model training was carried out using XGBoost, which internally implements early stopping by monitoring performance on a validation set during training. In addition to this, a separate hold-out validation set is used to double-check the model's behavior and to fine-tune hyperparameters via grid search.

Given the class imbalance in the target variable, the evaluation focuses primarily on the **F1-score**, which balances precision and recall. Other metrics, such as precision, recall, and AUC-ROC (Area Under the ROC Curve), were also computed to provide a more complete picture of model performance. Final results are reported on the independent test set, which remained unseen throughout the model development process, ensuring a robust and unbiased evaluation.

V. Results

A. Experimental Setup and Evaluation Strategy

To evaluate model performance and isolate the contribution of different feature sets, we trained and compared four configurations using the XGBoost algorithm:

TABLE I
MODEL PERFORMANCE COMPARISON BY FEATURE SET (EVALUATED ON THE TEST SET)

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
A	0.6406	0.6254	0.6454	0.6352	0.6408
B	0.6802	0.6605	0.7004	0.6799	0.6808
C	0.6736	0.6557	0.6878	0.6714	0.6740
D (Final)	**0.6910**	**0.6705**	**0.7130**	**0.6911**	**0.6916**

- **Model A (ATP Raw):** based solely on raw ATP attributes (e.g., rank, age, surface), without any feature engineering.
- **Model B (Betting Odds Raw):** included only implied probabilities derived from pre-match betting odds.
- **Model C (ATP Engineered):** featured only engineered ATP variables (e.g., momentum, style, context), but excluded betting odds.
- **Model D (ATP + Odds Engineered):** our final model, which combines both ATP features and betting odds with full feature engineering.

All models were trained on the same time-based training set and evaluated using identical validation and test splits.

B. Comparative Performance of Feature Sets

Table I confirms the value of ATP stats and betting odds:

- Model A offers a solid baseline, reaching 64.1% accuracy and an F1-score of 63.5%. This shows that even without advanced feature engineering, the cleaned ATP statistics carry meaningful predictive information.
- Model B performs better, with 68.0% accuracy and 67.9% F1-score. This confirms the predictive power of market-based expectations, but not dramatically above the raw ATP baseline.
- Model C reaches 67.4% accuracy. This demonstrates that careful data preparation can almost match the predictive power of the odds alone – a remarkable result, especially given that the model is now interpretable through tennis-specific logic.
- Model D achieves the best performance, with an accuracy of 69.1% and F1-score of 69.1%. This confirms that engineered ATP features and betting data are complementary, leading to a boost in performance.

It is interesting that, inasmuch as odds are a major component, the improvement they make toward our current ATP model is quite modest – with +2% as an increase in accuracy as well as F1-score. This finding supports one of the basic hypotheses of this work: *meticulous feature engineering, together with expertise in the subject matter, could rival the competence of the professionals in the marketplace*.

From a broader perspective, these results suggest that *the bottleneck to further improvements may lie more in data representation than in model architecture*. In fact, the current model, which relies entirely on XGBoost without deep learning or ensemble stacks, already approaches the state-of-the-art 72% accuracy benchmark reported in the literature [3], [4].

TABLE II
PERFORMANCE OF ML ALGORITHMS (MODEL C)

Algorithm	**Accuracy**	**F1-score**	**AUC-ROC**
Logistic Regression	0.6673	0.6618	0.6674
Random Forest	0.6654	0.6629	0.6658
LightGBM	**0.6743**	**0.6710**	**0.6746**
XGBoost	0.6715	0.6693	0.6720

TABLE III
PERFORMANCE OF ML ALGORITHMS (MODEL D)

Algorithm	**Accuracy**	**F1-score**	**AUC-ROC**
Logistic Regression	0.6730	0.6673	0.6731
Random Forest	0.6780	0.6750	0.6784
LightGBM	0.6871	0.6878	0.6878
XGBoost (Final)	**0.6913**	**0.6920**	**0.6920**

C. Comparison of ML Algorithms: Model C vs. Model D

To futher assess the impact of feature engineering as opposed to model complexity, we performed a comparison of four machine learning classifiers trained on the same ATP dataset under two configurations: Model C and Model D.

a) Model C: Table II presents the performance of XGBoost, Random Forest, LightGBM [19], and Logistic Regression using only ATP-engineered features. While LightGBM marginally outperformed the other models, all classifiers showed solid performancecs with F1-scores ranging from 0.661 to 0.671.

b) Model D: Table III summarizes results when the same models are trained using the full feature set (engineered ATP + betting odds). Here, XGBoost emerges as the most effective model, achieving the highest accuracy (0.6913), F1-score (0.6920), and AUC-ROC (0.6920). Figure 2 highlights that both models exhibit similar ROC curve patterns, indicating that their overall discriminative behavior remains largely consistent. These findings suggest that:

- Feature engineering on ATP data alone yields strong results, with XGBoost reaching an F1-score of 0.6693.
- The inclusion of betting odds improves performance across all models, with XGBoost benefitting the most from this additional information.
- Even in Model C, which excludes odds, well-prepared ATP features enable performance levels close to those obtained using expert-derived market signals.

D. Feature Importance Revisited

To better understand what drives the predictions of our model, we analyzed feature importance for the XGBoost classifier under two configurations: with and without betting odds.

a) Model D: Table IV shows the top 10 features in the full model. Notably, the most influential variables are directly derived from betting markets. Features such as `implied_prob_p2`, `odds_diff_prob`, and `odds_diff` dominate the list, indicating that bookmakers' expectations significantly contribute to prediction performance.

b) Model C: When odds-related variables are excluded, the model relies more heavily on engineered features from ATP data. Table V shows `rank_diff_log` as dominant, followed by other indicators such as `win_rate_diff`

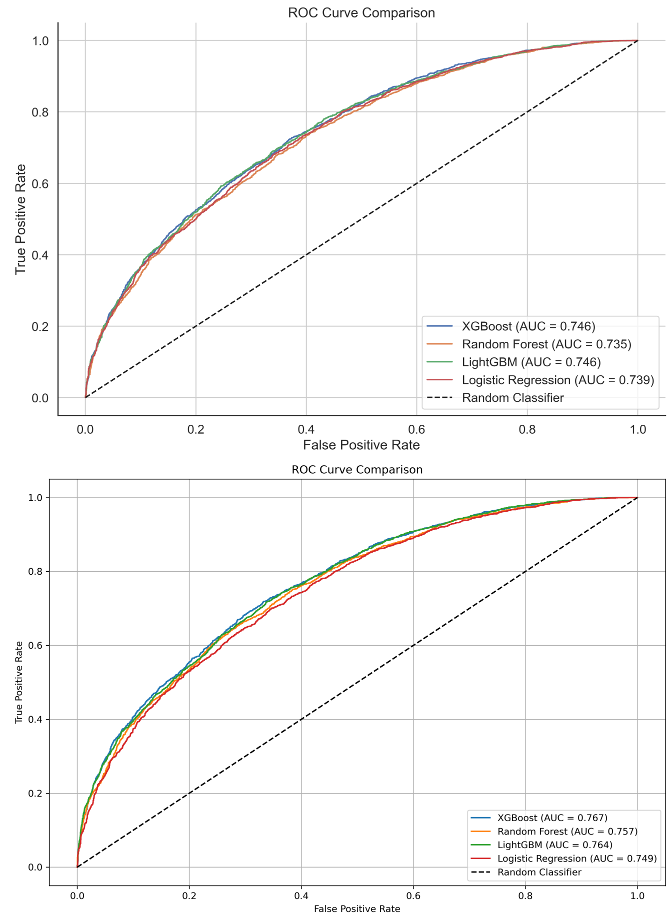

Fig. 2. **ROC curve comparison across ML models.** Top: Model C – LightGBM slightly outperforms other models in probabilistic classification. Bottom: Model D – XGBoost achieves the highest AUC (0.767), indicating superior discrimination. This AUC is calculated using predicted probabilities, unlike Tables II and III, which use class labels.

TABLE IV
TOP 10 MOST IMPORTANT FEATURES (ATP + BETTING ODDS)

Feature	**Importance Score**
implied_prob_p2	0.3424
odds_diff_prob	0.2683
odds_diff	0.0871
implied_prob_p1	0.0320
P2_Set_Finalization_Momentum	0.0145
best_of	0.0142
P1_Set_Finalization_Momentum	0.0134
tourney_importance	0.0134
match_spw	0.0131
P1_Odds_Momentum	0.0102

TABLE V
TOP FEATURES WHEN USING ATP-ONLY ENGINEERED DATA

Feature	Importance Score
rank_diff_log	0.2637
P1_Odds_Momentum	0.1326
win_rate_diff	0.1231
P2_Odds_Momentum	0.0999
win_rate_su_diff	0.0542
odds_momentum_diff	0.0343
P1_Odds_Rolling15	0.0335
P2_Odds_Rolling15	0.0319
best_of	0.0167
P2_Set_Finalization_Index_Long	0.0156
Style_Interaction_Advantage	0.0149
player_1_rank_log	0.0124

and `win_rate_su_diff`. Interestingly, even in this setting, `P1_Odds_Momentum` and related features like `P1_Odds_Rolling15` still appear. As previously mentioned, we are not using the current betting odds themselves, but rather statistical constructs derived from them – such as momentum indicators or rolling averages over the last 15 matches. These features do not capture the bookmaker's real-time estimate but rather provide a historical, smoothed approximation of perceived player performance over time.

E. Model Limitations

While the model demonstrates promising predictive performance, several constraints must be acknowledged:

- **Limited granularity:** The dataset lacks in-match evolution, such as set-by-set statistics or real-time odds fluctuations, limiting the model's ability to capture dynamic momentum shifts during the match.
- **No use of live/in-match stats:** Predictions are generated prior to match start and do not consider evolving factors such as injuries, fatigue, or in-play performance, which can drastically shift outcome probabilities during the match.

VI. CONCLUSIONS AND FUTURE WORK

This study examined the use of machine learning to predict professional tennis outcomes using only pre-match data. The XGBoost-based model achieved an accuracy of 69% in unseen matches, remarkably close to the 72% threshold of the best-performing approaches. A key contribution lies in the creation of informative features from ATP statistics, which, when combined with bookmaker odds, consistently outperformed models based on either data source alone.

Notably, Model C – based on historical data and domain-specific feature engineering – performed nearly on par with Model D (the history and odds model). This supports the idea that careful feature construction can rival expert-derived market signals. Variables like `rank_diff_log` and `win_rate_diff` were particularly influential, and LightGBM also yielded competitive results, reinforcing the strength of boosting-based approaches.

Future work includes integrating live betting data via APIs to capture match dynamics in real time and experimenting with alternative architectures (e.g., neural networks, hybrid models).

Overall, this study shows that a designed ML model – built on pre-match features and enriched with bookmaker odds – can match the predictive performance typically expected from professional betting markets. While not claiming superiority, the results suggest that combining statistical features with domain knowledge offers a competitive and interpretable alternative for forecasting tennis match outcomes.

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